STAT GU4204 — Practice Midterm Questions

Covers all material through Lecture Slide Set 6

Question types: Fill-in-the-blank, True/False, Multiple Choice, Listing, Computation

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Part A: Fill in the Blank

Question 1 (Fill in the Blank)

A function $T = \varphi(X_1, \ldots, X_n)$ that depends only on the observed data and contains no unknown parameters is called a _____.

Question 2 (Fill in the Blank)

An estimator $\hat{\theta}_n$ is said to be _____ if $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \to \infty$.

Question 3 (Fill in the Blank)

The decomposition $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{bias}(\hat{\theta})]^2$ is called the _____ decomposition.

Question 4 (Fill in the Blank)

A statistic T is _____ for θ if the conditional distribution of $\mathbf{X}$ given $T = t$ does not depend on θ.

Question 5 (Fill in the Blank)

The function $\dot{\ell}(x, \theta) = \frac{\partial}{\partial \theta} \log f(x, \theta)$ is called the _____, and its expected value equals _____.

Question 6 (Fill in the Blank)

A random variable $\Psi(\mathbf{X}_n, g(\theta))$ whose distribution does not depend on θ is called a _____.

Question 7 (Fill in the Blank)

If a prior distribution and its resulting posterior distribution belong to the same parametric family, the prior is called a _____ prior.

Question 8 (Fill in the Blank)

Under squared error loss, the Bayes estimator of θ is the _____ of the posterior distribution.

Question 9 (Fill in the Blank)

If $X_1, \ldots, X_n$ are iid $N(\mu, \sigma^2)$, then $\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\sigma^2}$ follows a _____ distribution.

Question 10 (Fill in the Blank)

The property that if $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$, is called _____ of the MLE.

Part B: True or False

Circle T or F. No justification required.

Question 11 (True/False)

T **F** If $Z_n \xrightarrow{d} Z$, then $Z_n \xrightarrow{P} Z$.

Question 12 (True/False)

T **F** The MLE is always an unbiased estimator of the parameter.

Question 13 (True/False)

T **F** If $\hat{\theta}_n$ is a consistent estimator of θ, then $\hat{\theta}_n$ is unbiased for all n.

Question 14 (True/False)

T **F** The Cramér-Rao lower bound applies to $X_1, \ldots, X_n$ iid Uniform$[0, \theta]$.

Question 15 (True/False)

T **F** For $X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$, $\bar{X}$ and s^2 are independent random variables.

Question 16 (True/False)

T F If $(2.3, 4.7)$ is a 95% confidence interval for μ, then μ lies in $(2.3, 4.7)$ with probability 0.95.

Question 17 (True/False)

T F The CLT guarantees that $\bar{X}_n$ follows an exact normal distribution for all n.

Question 18 (True/False)

T F A biased estimator can have lower MSE than any unbiased estimator of the same parameter.

Question 19 (True/False)

T F If $T(\mathbf{X})$ is sufficient for θ and $\delta(\mathbf{X})$ is any estimator of $g(\theta)$, then $E[\delta(\mathbf{X})|T]$ has MSE no larger than that of $\delta(\mathbf{X})$.

Question 20 (True/False)

T F The t_n distribution converges to $N(0, 1)$ as $n \to \infty$.

Question 21 (True/False)

T F For $X_1, \ldots, X_n$ iid from any distribution with finite variance, $\bar{X}_n$ is always the MVUE of μ.

Question 22 (True/False)

T F A sufficient statistic must be a single real number (cannot be vector-valued).

Question 23 (True/False)

T F As the sample size $n \to \infty$, the posterior distribution concentrates around the true θ regardless of the prior.

Question 24 (True/False)

T F For iid Uniform$[0, \theta]$ data, $2\bar{X}$ has lower MSE than $X_{(n)}$.

Part C: Multiple Choice

Circle the best answer.

Question 25 (Multiple Choice)

Suppose $X_1, \ldots, X_n$ are iid Poisson(λ). The MLE of λ is:

(a) $\frac{1}{n}\sum_{i=1}^{n} X_i^2$

(b) $\bar{X}$

(c) $\frac{1}{\bar{X}}$

(d) $\sqrt{\bar{X}}$

Question 26 (Multiple Choice)

Which of the following is a sufficient statistic for θ when $X_1, \ldots, X_n$ are iid Exp(θ)?

(a) $\max(X_1, \ldots, X_n)$

(b) $\min(X_1, \ldots, X_n)$

(c) $\sum_{i=1}^{n} X_i$

(d) $\bar{X}^2$

Question 27 (Multiple Choice)

Suppose $X_1, \ldots, X_{25}$ are iid $N(\mu, 9)$ with $\sigma^2 = 9$ known. A 95% CI for μ is:

(a) $\bar{X} \pm 1.96 \cdot \frac{3}{25}$

(b) $\bar{X} \pm 1.96 \cdot \frac{3}{\sqrt{25}}$

(c) $\bar{X} \pm 1.96 \cdot \frac{9}{\sqrt{25}}$

(d) $\bar{X} \pm t_{24,0.025} \cdot \frac{3}{\sqrt{25}}$

Question 28 (Multiple Choice)

The Fisher information $I(\theta)$ for a single observation from Bernoulli(θ) is:

(a) $\theta(1-\theta)$

(b) $\frac{1}{\theta}$

(c) $\frac{1}{\theta(1-\theta)}$

(d) $\frac{1}{\theta^2}$

Question 29 (Multiple Choice)

If $Z \sim N(0,1)$ and $V \sim \chi^2_n$ are independent, then $T = Z/\sqrt{V/n}$ follows:

(a) $N(0,1)$

(b) χ^2_{n+1}

(c) t_n

(d) $F_{1,n}$

Question 30 (Multiple Choice)

The Cramér-Rao inequality does NOT apply when:

(a) X_i iid $N(\mu, \sigma^2)$

(b) X_i iid Poisson(λ)

(c) X_i iid Uniform$[0, \theta]$

(d) X_i iid Exponential(θ)

Question 31 (Multiple Choice)

For Bayesian inference with $X_1, \ldots, X_n$ iid Bernoulli(θ) and prior $\theta \sim$ Beta(α, β), the posterior distribution of θ is:

(a) Beta$(\alpha + n, \beta)$

(b) Beta$(\alpha + \sum x_i, \beta + n - \sum x_i)$

(c) Gamma$(\alpha + \sum x_i, \beta + n)$

(d) $N(\bar{x}, 1/n)$

Part D: Listing / Short Answer

Question 32 (Listing)

List the three regularity conditions (A.1, A.2, A.3) required for the Cramér-Rao inequality.

Question 33 (Listing)

List three properties of the MLE. For each, state whether it always holds or requires conditions.

Question 34 (Short Answer)

State the Weak Law of Large Numbers. What are the conditions required?

Question 35 (Short Answer)

State the Central Limit Theorem precisely. What are the conditions?

Question 36 (Short Answer)

State the Factorization Criterion for sufficient statistics.

Question 37 (Short Answer)

State Proposition 6.3 (the three results about $\bar{X}$ and s^2 for a normal random sample).

Question 38 (Listing)

Name two estimation methods covered in this course and briefly describe each (one sentence each).

Question 39 (Short Answer)

Explain the difference between an *estimator* and an *estimate*. Give an example.

Question 40 (Short Answer)

What does it mean for a confidence interval to have "confidence level $1 - \alpha$"? Why is it incorrect to say "θ is in the interval with probability $1 - \alpha$"?

Part E: Computational Problems

Question 41 (MLE Computation)

Suppose $X_1, \ldots, X_n$ are iid Exponential(θ) with pdf $f(x|\theta) = \theta e^{-\theta x}$ for $x > 0$.

(a) Write down the likelihood function $L(\theta)$ and the log-likelihood $\ell(\theta)$.

(b) Find the MLE $\hat{\theta}$ of θ.

(c) Is $\hat{\theta}$ unbiased? Justify.

(d) Using the invariance property, find the MLE of $E(X_1) = 1/\theta$.

Question 42 (Method of Moments)

Suppose $X_1, \ldots, X_n$ are iid Gamma(α, β) with $E(X) = \alpha/\beta$ and $\text{Var}(X) = \alpha/\beta^2$.

(a) Find the method of moments estimators $\hat{\alpha}$ and $\hat{\beta}$.

(b) Are these estimators consistent? Explain why or why not.

Question 43 (Sufficient Statistics)

Let $X_1, \ldots, X_n$ be iid $N(\mu, \sigma^2)$ where both μ and σ^2 are unknown.

(a) Write the joint pdf $f(\mathbf{x}|\mu, \sigma^2)$.

(b) Use the Factorization Criterion to find a sufficient statistic for (μ, σ^2).

(c) Why is a single statistic (like $\sum X_i$) not sufficient in this case?

Question 44 (MSE Comparison)

Suppose $X_1, \ldots, X_n$ are iid Uniform$[0, \theta]$.

Consider two estimators of θ:

- $T_1 = 2\bar{X}$
- $T_2 = \left(1 + \frac{1}{n}\right) X_{(n)}$

(a) Show that both T_1 and T_2 are unbiased for θ.
Hint: $E(X_{(n)}) = \frac{n}{n+1}\theta$.

(b) Compute MSE(T_1).
Hint: $\text{Var}(X_i) = \theta^2/12$.

(c) Given that MSE$(T_2) = \frac{\theta^2}{n(n+2)}$, which estimator is better for $n = 5$? For $n = 50$?

Question 45 (Delta Method)

Let $X_1, \ldots, X_n$ be iid Bernoulli(θ) where $0 < \theta < 1$.

(a) State the asymptotic distribution of $\bar{X}_n$ (using the CLT).

(b) Using the delta method, find the asymptotic distribution of $g(\bar{X}_n) = \log\left(\frac{\bar{X}_n}{1-\bar{X}_n}\right)$ (the log-odds).
Hint: $g'(x) = \frac{1}{x(1-x)}$.

Question 46 (Confidence Interval Construction)

Suppose $X_1, \ldots, X_{16}$ are iid $N(\mu, \sigma^2)$ with σ^2 **unknown**. You observe $\bar{x} = 5.2$ and $s = 2.4$.

(a) What pivot do you use to construct a CI for μ? State its distribution.

(b) Construct a 95% confidence interval for μ.
You may use $t_{15,0.025} = 2.131$.

(c) How would your answer change if $\sigma = 2.4$ were *known* instead? Which interval is wider and why?

Question 47 (Fisher Information and CRLB)

Let $X_1, \ldots, X_n$ be iid Poisson(θ) with pmf $f(x|\theta) = e^{-\theta}\theta^x/x!$.

(a) Compute the score function $\dot{\ell}(x, \theta)$.

(b) Compute the Fisher information $I(\theta)$.

(c) State the Cramér-Rao lower bound for any unbiased estimator of θ.

(d) Is $\bar{X}$ the MVUE of θ? Justify.

Question 48 (Bayesian Inference)

Suppose $X_1, \ldots, X_n$ are iid Poisson(λ) and we use the prior $\lambda \sim \text{Gamma}(\alpha, \beta)$.

(a) Find the posterior distribution of $\lambda|\mathbf{x}$.

(b) Find the Bayes estimator of λ under squared error loss.

(c) Show that as $n \to \infty$, the Bayes estimator converges to the MLE.

Question 49 (Sampling Distributions)

Suppose $X_1, \ldots, X_9$ are iid $N(50, 16)$.

(a) What is the distribution of $\bar{X}$?

(b) What is the distribution of $\frac{\sum_{i=1}^{9}(X_i - \bar{X})^2}{16}$?

(c) What is the distribution of $\frac{\sqrt{9}(\bar{X} - 50)}{s}$?

(d) Compute $P(\bar{X} > 52)$.

Question 50 (Rao-Blackwell)

Let $X_1, \ldots, X_n$ be iid Bernoulli(θ).

(a) Show that $T = \sum_{i=1}^{n} X_i$ is sufficient for θ.

(b) Consider the crude estimator $\delta(X_1, \ldots, X_n) = X_1$ (just the first observation). Show it is unbiased for θ.

(c) Compute $E[X_1 | T = t]$ and verify it improves upon X_1.

Hint: Given $T = t$, each X_i is equally likely to be 1 or 0, so $E[X_1 | T = t] = t/n$.

Question 51 (Chi-Squared and CI for Variance)

Suppose $X_1, \ldots, X_{21}$ are iid $N(\mu, \sigma^2)$ and we observe $s^2 = 12.5$.

(a) What is the distribution of $(n-1)s^2/\sigma^2$?

(b) Construct a 90% confidence interval for σ^2.
You may use: $\chi^2_{20,0.05} = 31.41$ and $\chi^2_{20,0.95} = 10.85$.

Question 52 (Chebyshev / Sample Size)

A machine fills bottles with a mean of μ ounces and a standard deviation of $\sigma = 0.5$ ounces. You want to estimate μ by $\bar{X}_n$.

(a) Using Chebyshev's inequality, how large must n be so that $P(|\bar{X}_n - \mu| \geq 0.1) \leq 0.01$?

(b) Using the CLT approximation instead, how large must n be so that a 99% CI for μ has total width at most 0.2?
You may use $z_{0.005} = 2.576$.

Question 53 (Consistency)

Let $X_1, \ldots, X_n$ be iid with mean μ and variance $\sigma^2 < \infty$.

(a) Prove that $\bar{X}_n$ is a consistent estimator of μ using Chebyshev's inequality.

(b) Using the continuous mapping theorem, show that $\bar{X}_n^2$ is a consistent estimator of μ^2.

Question 54 (MLE for Uniform)

Suppose $X_1, \ldots, X_n$ are iid Uniform$[0, \theta]$.

(a) Write the likelihood $L(\theta)$.

(b) Explain why you cannot set $L'(\theta) = 0$ to find the MLE.

(c) Find the MLE of θ by reasoning about the shape of $L(\theta)$.

(d) Is the MLE unbiased? If not, find its bias.
Hint: $E(X_{(n)}) = \frac{n}{n+1}\theta$.

Question 55 (Normal MLE Bias)

Let $X_1, \ldots, X_n$ be iid $N(\mu, \sigma^2)$.

(a) The MLE of σ^2 is $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$. Show that $E(\hat{\sigma}^2) = \frac{n-1}{n}\sigma^2$.
Hint: Use the fact that $\sum(X_i - \bar{X})^2/\sigma^2 \sim \chi^2_{n-1}$.

(b) Is $\hat{\sigma}^2$ unbiased? What modification makes it unbiased?

(c) For $n = 10$, compute the bias of $\hat{\sigma}^2$.

Question 56 (CRLB with $g(\theta)$)

Let $X_1, \ldots, X_n$ be iid Bernoulli(θ). Suppose we want to estimate $g(\theta) = \theta^2$.

(a) Compute $g'(\theta)$.

(b) Using the Fisher information $I(\theta) = \frac{1}{\theta(1-\theta)}$, state the CRLB for any unbiased estimator of θ^2.

(c) Simplify the bound. Does $(\bar{X})^2$ achieve this bound?

Answer Key

Part A: Fill in the Blank

1. statistic
2. consistent
3. bias-variance
4. sufficient
5. score function; zero (0)
6. pivot
7. conjugate
8. mean (posterior mean)
9. χ^2_{n-1} (chi-squared with $n-1$ degrees of freedom)
10. invariance

Part B: True or False

11. **F** — conv in dist does NOT imply conv in prob (only if limit is constant)
12. **F** — MLE of σ^2 in Normal is biased; $X_{(n)}$ for Uniform is biased
13. **F** — consistency is a large-n property; estimator can be biased for each finite n
14. **F** — Uniform$[0, \theta]$ violates regularity condition A.1 (support depends on θ)
15. **T** — Proposition 6.3; this independence characterizes the normal distribution
16. **F** — θ is fixed; the probability statement is about the random interval before data is observed
17. **F** — CLT gives an approximate distribution for large n; exact normality requires X_i to be normal
18. **T** — e.g., $X_{(n)}$ for Uniform$[0, \theta]$ has lower MSE than any unbiased estimator like $2\bar{X}$
19. **T** — Rao-Blackwell theorem
20. **T** — by WLLN, $V/n \xrightarrow{P} 1$, so denominator $\to 1$ and $T \xrightarrow{d} N(0, 1)$
21. **F** — $\bar{X}_n$ is unbiased for μ always, but MVUE depends on the model (need CRLB or other argument)
22. **F** — e.g., $(\sum X_i, \sum X_i^2)$ is sufficient for $N(\mu, \sigma^2)$ with both unknown
23. **T** — posterior consistency; data overwhelms the prior
24. **F** — MSE$(2\bar{X}) = \theta^2/(3n)$ while MSE$(X_{(n)})$ is of order θ^2/n^2, much smaller

Part C: Multiple Choice

25. **(b)** $\bar{X}$
26. **(c)** $\sum X_i$
27. **(b)** $\bar{X} \pm 1.96 \cdot 3/\sqrt{25} = \bar{X} \pm 1.176$
28. **(c)** $I(\theta) = 1/[\theta(1-\theta)]$
29. **(c)** t_n
30. **(c)** Uniform$[0, \theta]$ — support depends on θ, violating A.1
31. **(b)** Beta$(\alpha + \sum x_i,\ \beta + n - \sum x_i)$

Part D: Listing / Short Answer

32. **A.1**: Support $\{x : f(x, \theta) > 0\}$ does not depend on θ. **A.2**: Can interchange $\partial/\partial\theta$ and $\int$. **A.3**: $\partial/\partial\theta \log f(x, \theta)$ exists and is finite for all x, θ.
33. **Invariance** (always holds): $g(\hat{\theta})$ is MLE of $g(\theta)$. **Consistency** (under regularity conditions): $\hat{\theta}_n \xrightarrow{P} \theta$. **May be biased / may not exist / may not be unique** (examples: Normal $\hat{\sigma}^2$ is biased; Uniform$[\theta, \theta+1]$ MLE is not unique).
34. WLLN: If $X_1, \ldots, X_n$ are iid with $E|X_1| < \infty$, then $\bar{X}_n \xrightarrow{P} E(X_1) = \mu$. Conditions: iid, finite mean.
35. CLT: If X_i iid with mean μ and $0 < \sigma^2 < \infty$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0, 1)$. Conditions: iid, finite variance.
36. Factorization: $T = r(\mathbf{X})$ is sufficient for θ iff $f(\mathbf{x}, \theta) = u(\mathbf{x}) \cdot v(r(\mathbf{x}), \theta)$ where u depends only on data and v depends on θ only through $r(\mathbf{x})$.
37. Prop 6.3: For X_i iid $N(\mu, \sigma^2)$: (1) $\bar{X} \sim N(\mu, \sigma^2/n)$; (2) $s^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$; (3) $\bar{X}$ and s^2 are independent.
38. MOM: Set sample moments equal to population moments, solve. MLE: Find θ maximizing the likelihood $L(\theta) = \prod f(X_i, \theta)$.
39. An estimator is a random variable (function of data); an estimate is the realized number. E.g., $\hat{\mu} = \bar{X}_n$ is the estimator; $\hat{\mu} = 3.7$ is an estimate.
40. A level $1-\alpha$ CI means that before observing data, the random interval contains $g(\theta)$ with probability $\geq 1-\alpha$. After data is observed, θ is fixed and the realized interval either contains it or doesn't — we cannot assign a probability.

Part E: Computational Problems

41. **(a)** $L(\theta) = \theta^n e^{-\theta \sum x_i}$, $\ell(\theta) = n \log\theta - \theta \sum x_i$. **(b)** $\ell'(\theta) = n/\theta - \sum x_i = 0 \Rightarrow \hat{\theta} = n/\sum x_i = 1/\bar{X}$. **(c)** $E(1/\bar{X}) \neq \theta$ in general (Jensen's inequality), so biased. **(d)** By invariance, MLE of $1/\theta$ is $1/\hat{\theta} = \bar{X}$.

42. **(a)** $E(X) = \alpha/\beta = \bar{X}$ and $E(X^2) - [E(X)]^2 = \alpha/\beta^2$, so $\alpha/\beta^2 = \hat{\mu}_2 - \bar{X}^2$. Solving: $\hat{\beta} = \bar{X}/(\hat{\mu}_2 - \bar{X}^2)$ and $\hat{\alpha} = \bar{X}^2/(\hat{\mu}_2 - \bar{X}^2)$. **(b)** Yes, by LLN $\bar{X} \xrightarrow{P} \alpha/\beta$ and $\hat{\mu}_2 \xrightarrow{P} E(X^2)$, then CMT.

43. **(a)** $f(\mathbf{x}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum(x_i - \mu)^2\right)$. **(b)** Expand: $\sum(x_i - \mu)^2 = \sum x_i^2 - 2\mu\sum x_i + n\mu^2$. So $f = \underbrace{1}_{u(\mathbf{x})} \cdot (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum x_i^2 - 2\mu\sum x_i + n\mu^2}{2\sigma^2}\right)$. Depends on data through $(\sum x_i, \sum x_i^2)$.
(c) Two unknown parameters require (at least) two-dimensional sufficient statistic.

44. **(a)** $E(T_1) = 2\mu = 2 \cdot \theta/2 = \theta$. $E(T_2) = (1 + 1/n) \cdot \frac{n}{n+1}\theta = \theta$. Both unbiased. **(b)** $\mathrm{Var}(T_1) = 4\mathrm{Var}(\bar{X}) = 4 \cdot \theta^2/(12n) = \theta^2/(3n)$. Since unbiased, MSE $= \theta^2/(3n)$. **(c)** $n = 5$: T_1 MSE $= \theta^2/15 \approx 0.067\theta^2$; T_2 MSE $= \theta^2/35 \approx 0.029\theta^2$. T_2 better. $n = 50$: T_1 MSE $= \theta^2/150$; T_2 MSE $= \theta^2/2600$. T_2 much better.

45. **(a)** $\sqrt{n}(\bar{X}_n - \theta)/\sqrt{\theta(1-\theta)} \xrightarrow{d} N(0,1)$, so $\bar{X}_n \sim_{\text{approx}} N(\theta, \theta(1-\theta)/n)$. **(b)** $g'(\theta) = 1/[\theta(1-\theta)]$. By delta method: $\sqrt{n}(g(\bar{X}_n) - g(\theta)) \xrightarrow{d} N\left(0, \frac{\theta(1-\theta)}{1} \cdot \frac{1}{\theta^2(1-\theta)^2}\right) = N\left(0, \frac{1}{\theta(1-\theta)}\right)$.

46. **(a)** Pivot: $\sqrt{n}(\bar{X} - \mu)/s \sim t_{n-1} = t_{15}$. **(b)** $5.2 \pm 2.131 \cdot 2.4/\sqrt{16} = 5.2 \pm 2.131 \cdot 0.6 = 5.2 \pm 1.279 = (3.921, 6.479)$. **(c)** If σ known: $5.2 \pm 1.96 \cdot 0.6 = 5.2 \pm 1.176 = (4.024, 6.376)$. The t-interval is wider because $t_{15,0.025} = 2.131 > z_{0.025} = 1.96$ (heavier tails compensate for estimating σ).

47. **(a)** $\ell(x, \theta) = -\theta + x\log\theta - \log(x!)$. $\dot{\ell} = -1 + x/\theta = (x - \theta)/\theta$. **(b)** $I(\theta) = E[(X - \theta)^2/\theta^2] = \mathrm{Var}(X)/\theta^2 = \theta/\theta^2 = 1/\theta$. **(c)** CRLB: $\mathrm{Var}(T) \geq 1/(n \cdot 1/\theta) = \theta/n$. **(d)** $\mathrm{Var}(\bar{X}) = \theta/n =$ CRLB. Since $\bar{X}$ is unbiased and achieves the bound, it is the MVUE.

48. **(a)** Posterior $\propto \prod \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda} \propto \lambda^{\alpha + \sum x_i - 1}e^{-(\beta+n)\lambda}$. This is Gamma$(\alpha + \sum x_i,\ \beta + n)$. **(b)** Posterior mean: $\frac{\alpha + \sum x_i}{\beta + n}$. **(c)** As $n \to \infty$: $\frac{\alpha + \sum x_i}{\beta + n} = \frac{\alpha/n + \bar{x}}{(\beta/n) + 1} \to \bar{x} =$ MLE.

49. **(a)** $\bar{X} \sim N(50, 16/9)$, i.e., $N(50, 16/9)$. **(b)** χ^2_8. **(c)** t_8. **(d)** $P(\bar{X} > 52) = P\left(Z > \frac{52-50}{4/3}\right) = P(Z > 1.5) = 1 - \Phi(1.5) \approx 0.0668$.

50. **(a)** Joint pmf: $\prod \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum x_i}(1-\theta)^{n - \sum x_i} = \underbrace{1}_{u(\mathbf{x})} \cdot v(\sum x_i, \theta)$. So $T = \sum X_i$ is sufficient. **(b)** $E(X_1) = \theta$, so unbiased. **(c)** $E[X_1 | T = t] = t/n = \bar{X}$. MSE$(\bar{X}) = \theta(1-\theta)/n \leq \theta(1-\theta) =$ MSE(X_1). Strict improvement for $n \geq 2$.

51. **(a)** $(n-1)s^2/\sigma^2 = 20s^2/\sigma^2 \sim \chi^2_{20}$. **(b)** $P\left(\chi^2_{20,0.95} \leq \frac{20s^2}{\sigma^2} \leq \chi^2_{20,0.05}\right) = 0.90$. Inverting: $\sigma^2 \in \left(\frac{20 \cdot 12.5}{31.41}, \frac{20 \cdot 12.5}{10.85}\right) = (7.96, 23.04)$.

52. **(a)** Need $\sigma^2/(n\varepsilon^2) \leq 0.01$: $0.25/(n \cdot 0.01) \leq 0.01 \Rightarrow n \geq 2500$. **(b)** Width $= 2 \cdot z_{0.005} \cdot \sigma/\sqrt{n} \leq 0.2$: $2(2.576)(0.5)/\sqrt{n} \leq 0.2 \Rightarrow \sqrt{n} \geq 12.88 \Rightarrow n \geq 166$.

53. **(a)** $P(|\bar{X}_n - \mu| > \varepsilon) \le \mathrm{Var}(\bar{X}_n)/\varepsilon^2 = \sigma^2/(n\varepsilon^2) \to 0$. **(b)** $g(x) = x^2$ is continuous. By CMT, $\bar{X}_n \xrightarrow{P} \mu \Rightarrow \bar{X}_n^2 \xrightarrow{P} \mu^2$.

54. **(a)** $L(\theta) = \prod_{i=1}^n \frac{1}{\theta} I(0 \le x_i \le \theta) = \theta^{-n} \cdot I(\theta \ge x_{(n)})$. **(b)** L is not differentiable at $\theta = x_{(n)}$ (discontinuity); for $\theta > x_{(n)}$, $L'(\theta) = -n\theta^{-(n+1)} < 0$ (always decreasing). **(c)** L is maximized at the smallest allowable θ, which is $\hat{\theta} = X_{(n)}$. **(d)** $E(\hat{\theta}) = \frac{n}{n+1}\theta \ne \theta$. Bias $= -\frac{\theta}{n+1}$ (underestimates θ).

55. **(a)** $E(\hat{\sigma}^2) = \frac{1}{n}E\big[\sum(X_i - \bar{X})^2\big] = \frac{\sigma^2}{n}E(\chi^2_{n-1}) = \frac{\sigma^2(n-1)}{n}$. **(b)** Biased; $E(\hat{\sigma}^2) = \frac{n-1}{n}\sigma^2 \ne \sigma^2$. Modification: $s^2 = \frac{n}{n-1}\hat{\sigma}^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$. **(c)** For $n = 10$: bias $= \frac{9}{10}\sigma^2 - \sigma^2 = -\sigma^2/10$.

56. **(a)** $g'(\theta) = 2\theta$. **(b)** CRLB $= \frac{(2\theta)^2}{n \cdot 1/[\theta(1-\theta)]} = \frac{4\theta^2 \cdot \theta(1-\theta)}{n} = \frac{4\theta^3(1-\theta)}{n}$. **(c)** $(\bar{X})^2$ is biased for θ^2 ($E(\bar{X}^2) = \theta^2 + \theta(1-\theta)/n \ne \theta^2$), so CRLB does not directly apply to it (it's not unbiased).

Part F: Workflow Drill Problems

Each problem is tagged with the workflow type [W1–W10] from the cheat sheet workflow guide.

Question 1 (Variance of Linear Combinations [W1])

Let X, Y, Z be random variables with $\mathrm{Var}(X) = 5$, $\mathrm{Var}(Y) = 3$, $\mathrm{Var}(Z) = 2$, $\mathrm{Cov}(X, Y) = 1$, $\mathrm{Cov}(X, Z) = -2$, $\mathrm{Cov}(Y, Z) = 0$.

(a) Find $\mathrm{Var}(2X - Y + 3Z)$.

(b) Find $\mathrm{Var}(X + Y - Z + 7)$.

Question 2 (Chebyshev Bound & Sample Size [W2])

Let $X_1, \ldots, X_n$ be iid with mean $\mu = 10$ and variance $\sigma^2 = 25$.

(a) Use Chebyshev's inequality to find an upper bound on $P(|\bar{X}_n - 10| \ge 3)$.

(b) How large must n be so that $P(|\bar{X}_n - 10| \le 1) \ge 0.95$?

Question 3 (Markov Inequality [W2])

Let X be a non-negative random variable with $E(X) = 4$.

(a) Find an upper bound on $P(X \ge 20)$.

(b) If additionally $E(|X - E(X)|^4) = 1000$, find an upper bound on $P(|X - E(X)| \ge 5)$ using a 4th-moment bound.

Question 4 (Standard Delta Method [W3])

Let $X_1, \ldots, X_n$ be iid with mean θ and variance σ^2. Find the asymptotic distribution of $(\bar{X}_n)^4$.

Question 5 (Delta Method — Transformed Variance [W3])

Let $Y_n = \frac{1}{n}\sum_{i=1}^{n} X_i^2$, where $E(X^2) = \sigma^2$ and $\mathrm{Var}(X^2) = 2\sigma^4$. Find the asymptotic distribution of $\sqrt{Y_n}$.

Question 6 (Order Statistics [W4])

Let $X_1, \ldots, X_n$ be iid Unif$[0, \theta]$. Let $Y_n = \max(X_i)$.

(a) Find the CDF of Y_n.

(b) Find the CDF of $Z_n = n(Y_n - \theta)$ and determine its limiting distribution as $n \to \infty$.

Question 7 (MLE — Calculus Path [W5])

Let $X_1, \ldots, X_n$ be iid with PDF $f(x|\theta) = \theta x^{\theta-1}$ for $0 < x < 1$, $\theta > 0$. Find the MLE of θ.

Question 8 (MLE — Restricted Domain [W5])

Let $X_1, \ldots, X_{70}$ be iid Bernoulli(p). You observe $\sum_{i=1}^{70} x_i = 58$. Find the MLE of p if the parameter space is restricted to $p \in [1/2, 2/3]$.

Question 9 (MLE Invariance [W6])

Let $X_1, \ldots, X_n$ be iid Exponential(β). The MLE of β is $\hat{\beta} = 1/\bar{X}$.

(a) Find the MLE of the median of the distribution.

(b) Find the MLE of $P(X > 2)$.

Question 10 (MOM Estimator [W7])

Let $X_1, \ldots, X_n$ be iid Bernoulli(θ). Find the method of moments estimator of θ and verify it equals the MLE.

Question 11 (Consistency Proof [W8])

Let $X_1, \ldots, X_n$ be iid Exponential(β) with mean $1/\beta$. Show that $\hat{\beta} = 1/\bar{X}_n$ is consistent for β.

Question 12 (Gamma Integral [W9])

Evaluate $\int_0^\infty x^5 e^{-3x}\,dx$.

Question 13 (Expected Power of Gamma [W9])

Let $X \sim \text{Gamma}(2,3)$. Compute $E[X^4]$.

Question 14 (Sum of Exponentials [W10])

Let $X_1, \ldots, X_8$ be iid Exponential(β). Find the distribution of $S = X_1 + \cdots + X_8$.

Question 15 (Building a t-statistic [W10])

Let $X_1, \ldots, X_5$ be iid $N(0,1)$. Define $W = X_1/\sqrt{(X_2^2 + X_3^2 + X_4^2)/3}$.

(a) What is the distribution of W?

(b) What is the distribution of W^2?

Workflow Drill Answers

1. **(a)** $4(5)+1(3)+9(2)+2(2)(-1)(1)+2(2)(3)(-2)+2(-1)(3)(0) = 20+3+18-4-24+0 = \boxed{13}$.
   (b) $1(5)+1(3)+1(2)+2(1)(1)(1)+2(1)(-1)(-2)+2(1)(-1)(0) = 5+3+2+2+4+0 = \boxed{16}$

2. **(a)** $P(|\bar{X} - 10| \geq 3) \leq \text{Var}(\bar{X})/9 = 25/(9n) = \boxed{25/(9n)}$. **(b)** Need $25/(n \cdot 1) \leq 0.05$, so $\boxed{n \geq 500}$.

3. **(a)** Markov: $P(X \geq 20) \leq E(X)/20 = 4/20 = \boxed{1/5}$. **(b)** $P(|X - \mu| \geq 5) \leq \beta_4/5^4 = 1000/625 = 1.6$. Bound exceeds 1, so trivially ≤ 1 (uninformative).

4. $g(x) = x^4$, $g'(\theta) = 4\theta^3$. By delta method: $\boxed{N(\theta^4,\ 16\sigma^2\theta^6/n)}$.

5. $g(x) = \sqrt{x}$, $g'(\sigma^2) = 1/(2\sigma)$. Asymptotic variance: $(2\sigma^4/n) \cdot 1/(4\sigma^2) = \sigma^2/(2n)$. Answer: $\boxed{N(\sigma,\ \sigma^2/(2n))}$.

6. **(a)** $P(Y_n \leq y) = (y/\theta)^n$ for $0 < y < \theta$. **(b)** $P(Z_n \leq z) = (1 + z/(n\theta))^n \to \boxed{e^{z/\theta}}$ for $z \leq 0$.

7. $\ell(\theta) = n \log\theta + (\theta - 1)\sum \log x_i$. Setting $\ell'(\theta) = n/\theta + \sum \log x_i = 0$: $\boxed{\hat{\theta} = -n/\sum \log x_i}$.
   Check: $\ell'' = -n/\theta^2 < 0$. Note $\hat{\theta} > 0$ since $\log x_i < 0$.

8. Unrestricted MLE: $\hat{p} = 58/70 \approx 0.829 > 2/3$. Falls outside domain. $L(p) = p^{58}(1-p)^{12}$ is increasing on $[1/2, 2/3]$ (since maximum is to the right). So $\boxed{\hat{p} = 2/3}$.

9. **(a)** Median of $\text{Exp}(\beta)$ is $(\log 2)/\beta$. By MLE invariance: $\hat{m} = (\log 2)/\hat{\beta} = \boxed{(\log 2) \cdot \bar{X}}$. **(b)** $P(X > 2) = e^{-2\beta}$. By invariance: $\boxed{e^{-2/\bar{X}}}$.

10. MOM: $E(X) = \theta$, so $\hat{\theta}_{\text{MOM}} = \bar{X}$. MLE: $\ell = \sum x_i \log\theta + (n - \sum x_i)\log(1-\theta)$. Setting $\ell' = 0$: $\hat{\theta}_{\text{MLE}} = \bar{X}$. $\boxed{\text{MOM} = \text{MLE} = \bar{X}}$.

11. By WLLN: $\bar{X}_n \xrightarrow{P} E(X) = 1/\beta$. Let $g(x) = 1/x$; g is continuous at $1/\beta > 0$. By CMT: $\boxed{1/\bar{X}_n \xrightarrow{P} \beta}$.

12. Match to $\int_0^\infty x^{\alpha-1}e^{-\beta x}dx$ with $\alpha = 6$, $\beta = 3$. Answer: $\Gamma(6)/3^6 = 120/729 = \boxed{40/243}$.

13. $E[X^4] = \Gamma(2+4)/\big(3^4 \cdot \Gamma(2)\big) = \Gamma(6)/(81 \cdot 1) = 120/81 = \boxed{40/27}$.

14. MGF of S: $\prod_{i=1}^8 \frac{\beta}{\beta - t} = \left(\frac{\beta}{\beta-t}\right)^8$. This is the MGF of $\boxed{\text{Gamma}(8, \beta)}$.

15. **(a)** $X_1 \sim N(0,1)$ and $V = X_2^2 + X_3^2 + X_4^2 \sim \chi_3^2$, independent of X_1. So $W = X_1/\sqrt{V/3} \sim \boxed{t_3}$. **(b)** W^2 has an $F(1,3)$ distribution (square of a t_3).